

Substitution

Example

If $a = 4$, $b = 2$ and $c = 3$, what are the values of the following expressions?

- a. $a + b$
- b. $b - c$
- c. $ab + c$

When we substitute, we replace something or someone with another thing or person. For example, during a football match, the coach may substitute a player by swapping them out with another already on the field.

For our substitution here, we need to substitute a **letter** for a **number**.

Step One

When we substitute, we replace the letter for its value.

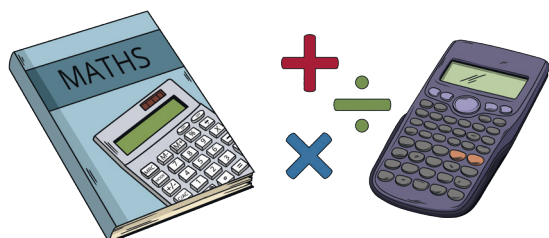
We are told that, in this example, a has a value of 4. Therefore, we can substitute a for 4.

- a. $4 + b$
- b. $b - c$
- c. $4b + c$

Step Three

Substitute the letter c for its value.

- a. $4 + 2$
- b. $2 - 3$
- c. $4 \times 2 + 3$ or $8 + 3$



Step Two

Now substitute the letter b for its value. b has a value of 2:

- a. $4 + 2$
- b. $2 - c$
- c. $4b$ is slightly different; we cannot simply place the 2 next to the 4,

i.e. the answer is not $42 + c$.

Rather, we must remember there is an invisible multiplication sign between the 4 and b .

So, our substitution becomes 4×2 .

$$4 \times 2 + c$$

which is equal to

$$8 + c$$

Step Four

You have substituted the values for the letters you have been given.

- a. $4 + 2 = 6$
- b. $2 - 3 = -1$
- c. $8 + 3 = 11$